

Four tetrahedral spanning trees versus a fourth derivative of 24

Disproof of Conjecture 00000004222

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Abstract

The boundary of a tetrahedron has four two-dimensional spanning trees. Its top-dimensional Laplacian acts on four faces, so its monic characteristic polynomial has fourth derivative $4! = 24$ everywhere, including at 1. This disproves the stated equality.

Statement and conventions

The conjecture asserts that a higher-dimensional tree count equals the derivative at 1 of a Laplacian characteristic polynomial, with derivative order equal to the number of faces in that dimension. We use the ordinary derivative of the characteristic polynomial $\det(tI - L)$, and the unweighted combinatorial Laplacian on that dimension's chains. No rescaling or reduced characteristic polynomial is specified in the source.

For a two-dimensional complex with complete one-skeleton, a spanning two-tree keeps all vertices and edges and selects triangular faces such that the selected boundary map is injective over $\mathbb{Q}$ and its image is $\ker \partial_1$. These are the chain descriptions of vanishing rational H_2 and H_1 . This agrees with the usual simplicial spanning-tree conditions here; see Duval–Klivans–Martin, Section 1.2.¹ We also check below that every tree in this example has trivial integral H_1 , so a squared torsion-order weight does not change the count.

The actual chain complex

Let $K = \partial[0, 1, 2, 3]$, the family of all proper subsets of four vertices. Its six increasing oriented edges and four increasing oriented faces are, respectively,

$$(01, 02, 03, 12, 13, 23), \quad (123, 023, 013, 012).$$

Use the usual boundaries $\partial_1[u, v] = [v] - [u]$ and $\partial_2[a, b, c] = [b, c] - [a, c] + [a, b]$. Their matrices are

$$B_1 = \begin{pmatrix} -1 & -1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Direct calculation gives $B_1 B_2 = 0$. Set $\sigma = (1, -1, 1, -1)^T$. Solving $B_2 x = 0$ gives

$$x_2 + x_3 = 0, \quad x_1 - x_3 = 0, \quad x_0 + x_3 = 0, \quad x = x_0 \sigma.$$

Thus $\ker B_2 = \mathbb{Q}\sigma$, and every nonzero two-cycle uses all four faces. For any one-cycle $z = (z_0, \dots, z_5)^T$, the first three rows of $B_1 z = 0$ imply

$$z_2 = -z_0 - z_1, \quad z_4 = z_0 - z_3, \quad z_5 = z_1 + z_3.$$

Consequently $a = (z_3, z_1, z_0, 0)^T$ satisfies $B_2 a = z$.

¹A. M. Duval, C. J. Klivans, and J. L. Martin, *Simplicial matrix-tree theorems*, Section 1.2.

Lemma 1. *A subset of the four triangular faces is a spanning two-tree precisely when it omits exactly one face. Therefore $\tau_2(K) = 4$.*

Proof. If face v is omitted, every supported two-cycle is a scalar multiple of σ with its v -coordinate zero, hence is zero. For a one-cycle z , take the filling a above and define

$$x = a - (a_v \sigma_v) \sigma.$$

Since $\sigma_v^2 = 1$, this has $x_v = 0$, while $B_2 x = B_2 a = z$. Thus omitting one face gives a spanning two-tree.

Selecting every face fails injectivity because $\sigma \neq 0$ is a cycle. If distinct faces v, w are omitted, let e_v denote the v th coordinate vector. The one-cycle $B_2 e_v$ cannot have a supported filling x : otherwise $B_2(x - e_v) = 0$, so $x - e_v = c\sigma$. Its w -coordinate forces $c = 0$, whereas its v -coordinate is -1 , a contradiction. This proves the classification and count. $\square$

All displayed filling formulas have integer coefficients. For integral z , both a and x are integral, so every integral one-cycle is a boundary in each of the four trees. The same kernel argument rules out integral two-cycles. In particular, $H_1(T; \mathbb{Z}) = 0$ for every counted tree T , and the usual weight $|\text{Tor } H_1(T; \mathbb{Z})|^2$ equals 1. Geometrically, these four complexes are also the four closed disks obtained by removing one open face of the tetrahedral sphere.

The prescribed derivative

There are no three-dimensional faces, so the top-dimensional Hodge Laplacian is exactly

$$L_2 = B_2^T B_2 \in M_4(\mathbb{Q}), \quad p(t) = \det(tI_4 - L_2).$$

Every 4×4 matrix has a monic characteristic polynomial of degree 4. Hence $p(t) = t^4 + c_3 t^3 + c_2 t^2 + c_1 t + c_0$, and

$$\boxed{\tau_2(K) = 4 \neq 24 = p^{(4)}(1), \quad f_2(K) = 4.}$$

One also commonly uses the up-down Laplacian $L_1^{\text{up}} = B_2 B_2^T$ on edges, as in the cited simplicial matrix-tree theorem. This convention also contradicts the claimed derivative order. Indeed, multiplication of the displayed matrix gives

$$B_2^T B_2 = 4I_4 - \sigma \sigma^T, \quad \sigma^T \sigma = 4.$$

Thus its eigenvalues are 4, 4, 4, 0. The 6×6 matrix $B_2 B_2^T$ has the same nonzero eigenvalues and three zero eigenvalues. Therefore

$$p_{\text{up}}(t) = t^3(t - 4)^3 = t^6 - 12t^5 + 48t^4 - 64t^3, \quad p_{\text{up}}^{(4)}(1) = 360 - 1440 + 1152 = 72 \neq 4.$$

Here the derivative order remains the number of two-faces, namely 4. This supplementary convention check is a paper calculation; the Lean proof below formalizes the 4×4 top-dimensional Laplacian.

The contradiction concerns the derivative and derivative order stated in the source. It does not invoke, or refute, a distinct identity involving reduced Laplacians, normalized coefficients, or products of nonzero eigenvalues. Even division by $4!$ would give 1, not the count 4.

Correspondence with Lean

The finite family `tetrahedron` is proved downward closed. The actual two-faces are enumerated bijectively by `faceSet`; the six increasing edges are enumerated by `tail` and `head`. The matrices `B_1` and `B_2` (written with Unicode subscripts in the source) are defined from the displayed

simplicial boundary formulas, before their entries are computed. Their linear maps satisfy `boundary_boundary`.

`IsSpanningTwoTree` uses those actual rational boundary maps and the support of a selected-face chain. The theorem `tree_iff_one_missing` proves the classification; the finite cardinality `treeCount` is then proved equal to 4. The object `characteristic` is Mathlib's actual `Matrix.charpoly` of `L_2`. `fourth_derivative` uses monicity, the polynomial degree, and the coefficient formula for iterated derivatives. The final theorem `tree_count_ne_prescribed_derivative` substitutes the actual cardinality of the face type as the derivative order. The formal proof is over $\mathbb{Q}$; the integral-homology and topological interpretations above are supplied by the explicit integer filling argument on paper, rather than claimed as imported Lean homology results.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build` and `lake env lean -DwarningAsError=true Main.lean`. On a new machine, `lake exe cache get` retrieves the official pinned dependency artifacts. The submission is checked in a fresh build directory. Printed dependencies use only the standard `propext`, `Classical.choice`, and `Quot.sound`. No additional axioms, proof placeholders, or numerical oracles are used. Run `tectonic main.tex` to reproduce the PDF; no auxiliary numerical script is needed. Logs, hashes, and visual-review records are included under `verification/`.

Conjecture source

The Last Math Competition, Conjecture 00000004222. The exact bilingual source and its commit and SHA-256 are preserved in this submission.